Model 1: Coupled Vegetation–Water–Sand Dynamics

Yutian–Minfeng section of the Taklamakan edge-locking belt

2026 CQU_MCM Problem B, Task 1 · first version, 2026-11-10

What this is. A daily model of one land cell of the belt. The two main states are vegetation cover V and effective soil moisture M; sand enters as a wind-driven flux that couples to both. All numbers below are produced by `Code/m1_run.py` and stored in `Results/model1/`. Evidence and page references are in `Docs/20261110_audit_check_and_model1.md`.

1 Data

Section. Southern margin between the Yutian and Minfeng oases, stylised as 100 km × 1.5 km = 15,000 ha. This is our assumption; no public boundary of the belt exists.

Data (2001–2024)	Source	Used for
NDVI and pixel reliability, 250 m, 16-day	MODIS MOD13Q1 v061	observed cover
Daily $T_{\max}$, $T_{\min}$, dew point, wind, radiation, pressure, rain	ERA5 / ERA5-Land (model fixed)	reference evapotranspiration, water balance
Hourly 10 m wind	ERA5	drift potential
Daily weather	NASA POWER	independent check of ET_0

Cover from NDVI. End-members are read from the section's own desert and oasis pixels:

$$\mathrm{FVC} = \min\Big\{1, \max\Big[0, \frac{\mathrm{NDVI} - 0.0505}{0.5827 - 0.0505}\Big]\Big\}. \tag{1}$$

Land classes (9×9-pixel windows chosen by rule from mean summer NDVI of 2001–03 and 2022–24): 218 *converted* pixels (bare, then vegetated: irrigated, treated land), 242 natural *ecotone* pixels, 162 *desert* pixels, 158 *oasis* pixels.

Climate of the section. Rain 31 mm/yr; ET_0 1663 mm/yr (FAO-56 Penman–Monteith; two independent sources agree within about 5%); mean temperature 13.8 °C; drift potential 45–55 VU; 450–590 h/yr with wind ≥ 6 m/s.

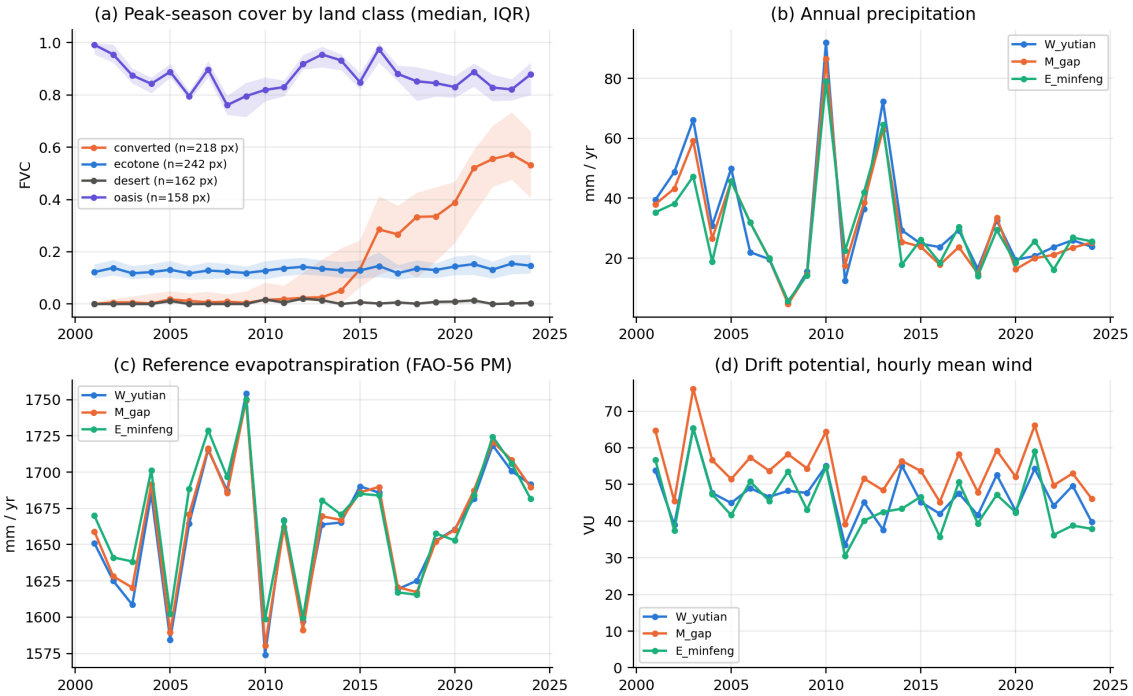


Figure 1: Observed cover by land class and yearly drivers on the section.

2 Formulas

Effective soil moisture. D_r is the root-zone depletion below field capacity (mm):

$$\mathrm{TAW} = 1000\, Z_r(\theta_{fc} - \theta_{wp}) = 75 \text{ mm}, \qquad M = 1 - \frac{D_r}{\mathrm{TAW}} \in [0, 1]. \tag{2}$$

Water balance (FAO-56 dual crop coefficient, daily, all terms in mm/day):

$$D_r(t+1) = D_r(t) + T + E - P - \eta I - G \quad \text{(water above field capacity drains)}, \tag{3}$$

$$K_s = \min\left\{1, \frac{M}{1-p}\right\}, \tag{4}$$

$$T = K_s\, K_{cb,\mathrm{full}}\, \min(1, 2V)\, \varphi(T_a)\, \mathrm{ET}_0, \tag{5}$$

$$E = \sum_x K_r(D_{e,x})\, f_{ew,x}\, K_{c,\max}\, \mathrm{ET}_0, \qquad \sum_x f_{ew,x} = 1 - V, \tag{6}$$

$$\varphi(T_a) = \min\left\{1, \max\left[0, \frac{T_a - 10}{25 - 10}\right]\right\}. \tag{7}$$

P is rain, I gross irrigation, η the share reaching the roots, G groundwater supply (zero under dunes), K_s the water-stress coefficient, T transpiration, E soil evaporation from the unshaded fraction, φ thermal activity from the 15-day mean air temperature.

Vegetation cover.

$$\frac{dV}{dt} = \underbrace{r\,\varphi\,K_s\,V\left(1 - \frac{V}{V_{\max}}\right)}_{\text{growth}} - \Big[\underbrace{m_0}_{\text{background}} + \underbrace{m_w\,\varphi\,(1 - K_s)}_{\text{drought}} + \underbrace{\varepsilon\, g}_{\text{sand}} + \underbrace{\gamma\,\mathrm{DP}^{2/3}}_{\text{wind drag}}\Big] V + b(t). \tag{8}$$

$b(t)$ is cover added by planting (the link to Task 2).

Sand. With U the hourly 10 m wind in knots and $U_t = 12$ knots (≈ 6 m/s):

$$\mathrm{DP} = \left\langle U^2 \, (U - U_t)_+ \right\rangle \quad \text{[vector units, VU]}, \tag{9}$$

$$g = \tfrac{1}{2} \, \mathrm{DP} \left[\tanh\big(d \, (v_c - V)\big) + 1 \right] (1 - V), \tag{10}$$

$$U_t^{\mathrm{wet}} = U_t \sqrt{1 + 1.21 \, (w - w')^{0.68}} \quad (w > w'). \tag{11}$$

Sand transport is switched off once cover passes the critical value v_c; a moist surface raises the threshold (w: gravimetric surface moisture in %).

Couplings. Water → vegetation (K_s); vegetation → water (transpiration, shading); vegetation → sand (g); sand → vegetation (εg, γ term); water → sand (wet threshold).

3 Parameters

Parameter	Value	Origin
θ_{fc}, θ_{wp}	0.12, 0.045 $\mathrm{m^3/m^3}$	FAO-56 Table 19, sand
$K_{cb,\mathrm{full}}$, $K_{c,\max}$	1.0, 1.2	FAO-56; $K_{cb,\mathrm{full}}$ uncertain (0.7–1.2)
Z_r, p, η	1.0 m, 0.6, 0.9	assumptions
r	1.43 ± 0.28 $\mathrm{yr^{-1}}$	fitted to converted pixels
$V_{\max}$	0.55 ± 0.03	fitted to converted pixels
m_0	0.02 $\mathrm{yr^{-1}}$	Yizhaq & Ashkenazy (2021)
m_w	1.0 $\mathrm{yr^{-1}}$	assumption (0.3–3); the weakest number
ε, γ, v_c, d	0.001, 8×10^{-4}, 0.30, 15	Yizhaq & Ashkenazy (2021)
Irrigation timing	every 15 d Mar–Oct, 10 d Jul–Aug	Tarim Desert Highway shelterbelt practice

4 Fit and hold-out

Fit	Calibration	Hold-out	Reading
Irrigated establishment (3 windows fitted, 1 held out)	$R^2 = 0.40$	$R^2 = 0.29$ (plain logistic 0.29; mean -0.02)	average rise captured; no better than a logistic because water is not limiting in these runs
Natural ecotone (fit ≤2016, test 2017–24)	RMSE 0.008	$R^2 = -0.76$ (same as the mean)	level 0.13 reproduced with $G = 166$ mm/yr; year-to-year change not explained
Bare desert (no fit)	–	cover → 0.001; observed 0.002	rain alone cannot hold cover

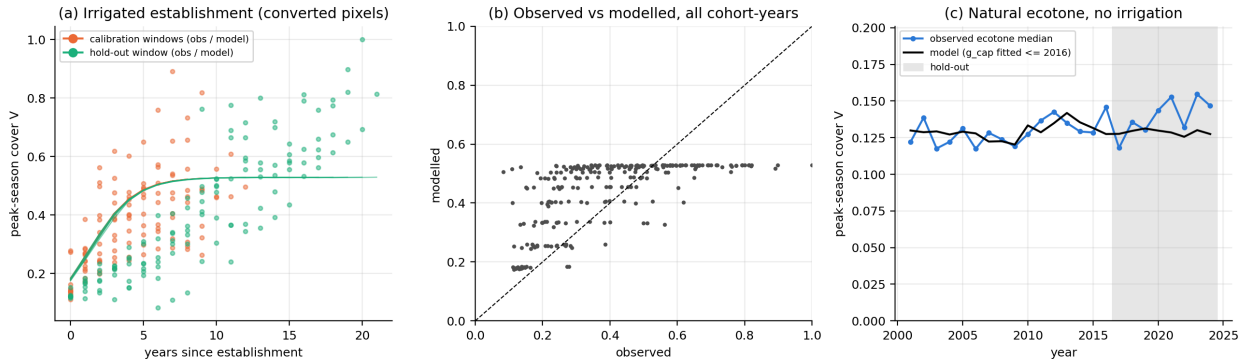


Figure 2: Calibration and hold-out.

5 Results

Gross irrigation (mm/yr)	0	100	200	300	400	500	600	800
Steady cover V	0.004	0.03	0.11	0.17	0.22	0.26	0.29	0.35
Sand transport / bare sand	0.98	0.95	0.88	0.80	0.70	0.55	0.36	0.10
Transpiration (mm/yr)	3	37	128	219	302	370	436	564
Soil evaporation (mm/yr)	21	75	74	72	71	71	70	68

- **Water constraint.** Holding 30% cover needs about 624 mm/yr (6,240 m^3/ha/yr; 94 million m^3/yr for 15,000 ha). Use the range 400–750 mm/yr: m_w and $K_{cb,\text{full}}$ are not fixed by data.
- **Establishment.** At 600 mm/yr a new planting reaches 30% cover in 13 years; at 400 mm/yr or less it never does.
- **Stopping irrigation.** Half the cover is lost in 2 years.
- **Water balance** closes to 10^{-12} mm over 24 years.

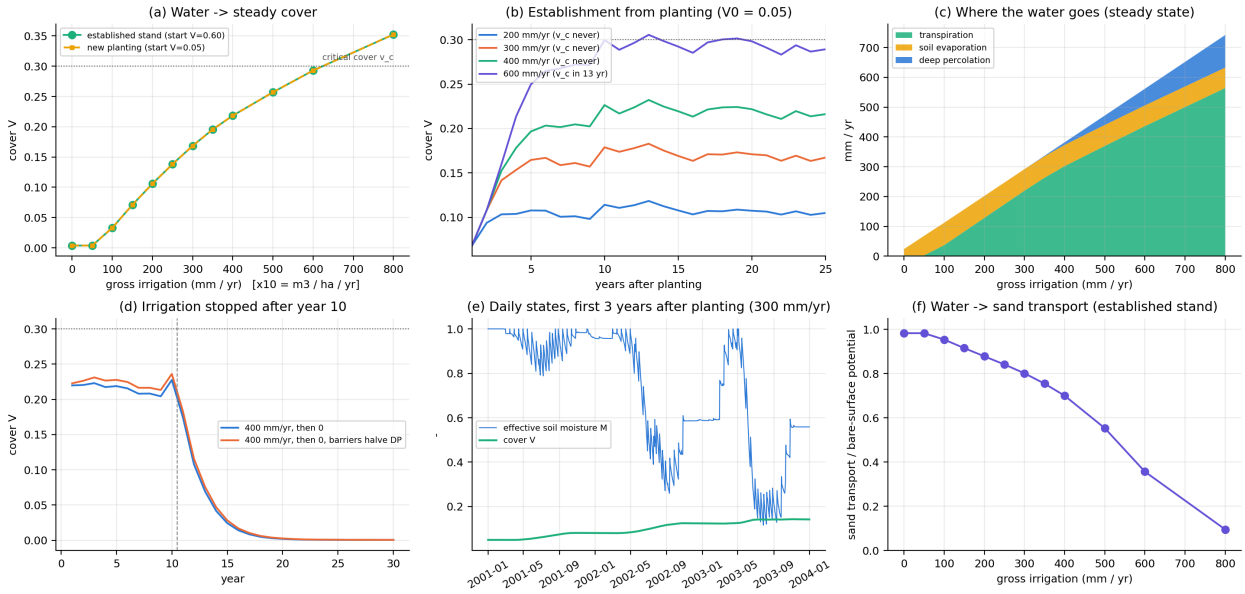


Figure 3: Model behaviour: water–cover curve, establishment, water use, abandonment, daily states, sand transport.

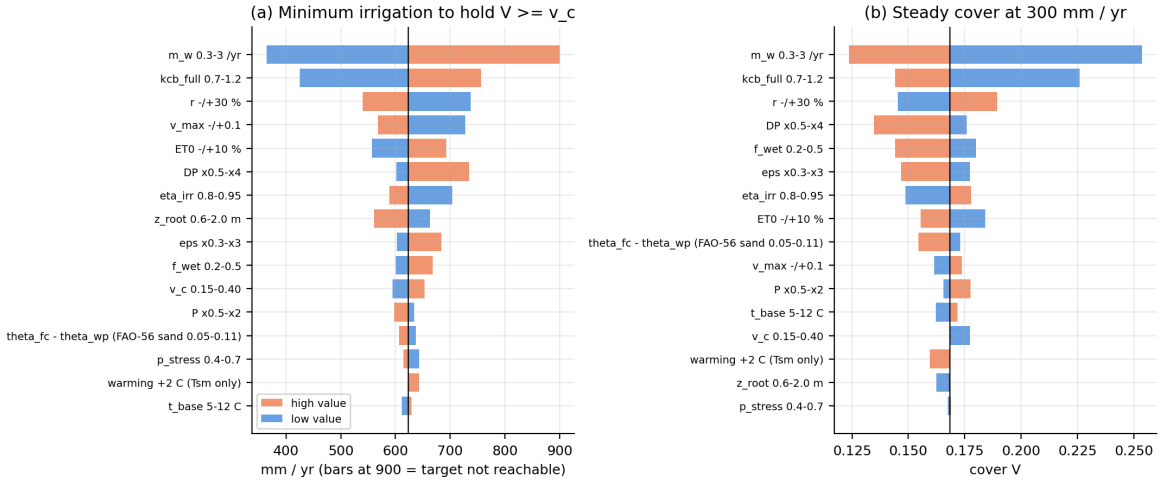


Figure 4: One-at-a-time sensitivity.

6 Limits

1. The water–cover curve is model output; no field measurement of belt cover against applied water is available to validate it.
2. Converted pixels are irrigated land of unknown type (shelter forest, orchard or field).
3. Year-to-year variation of natural cover is not explained.
4. Drift potential is relative (VU); converting it to a mass flux needs a local coefficient. ERA5 hourly mean wind misses sub-hourly peaks.
5. Salinity of irrigation water is not modelled; $v_c = 0.30$ comes from a model calibrated elsewhere.